# Sri Lanka Institute of Information Technology

IT2130 – Operating Systems and System Administration
Year 2, Semester 2- 2026

Practical Answer Submission

Practical Sheet No: 02

| | |
|---|---|
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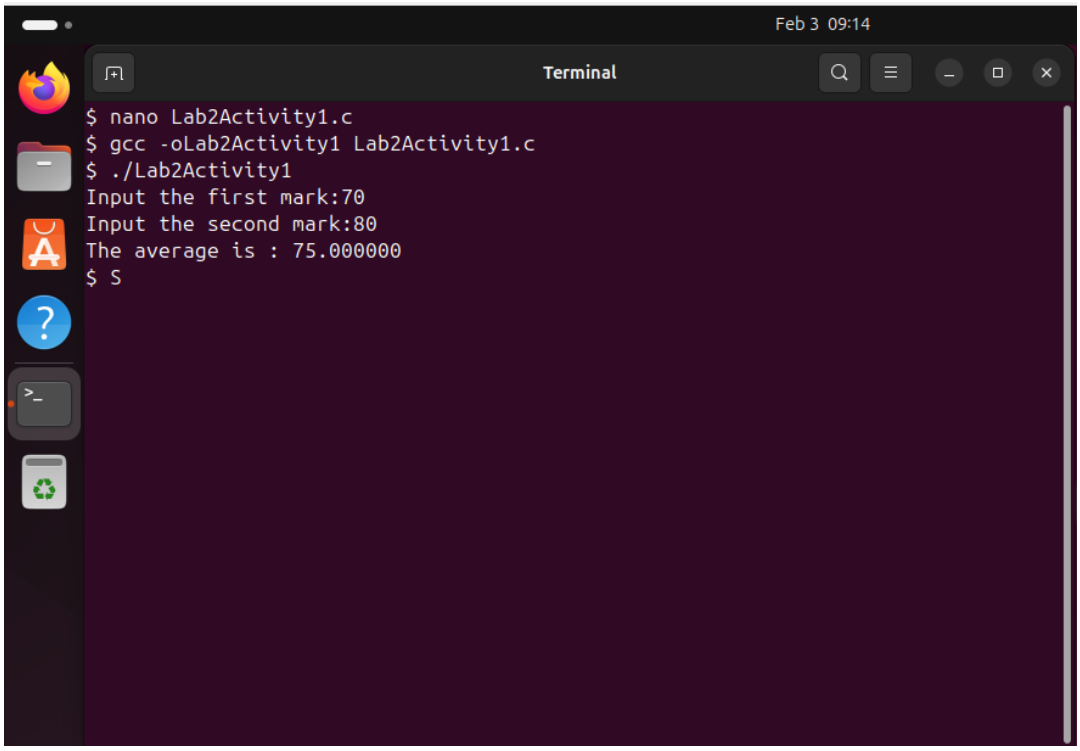
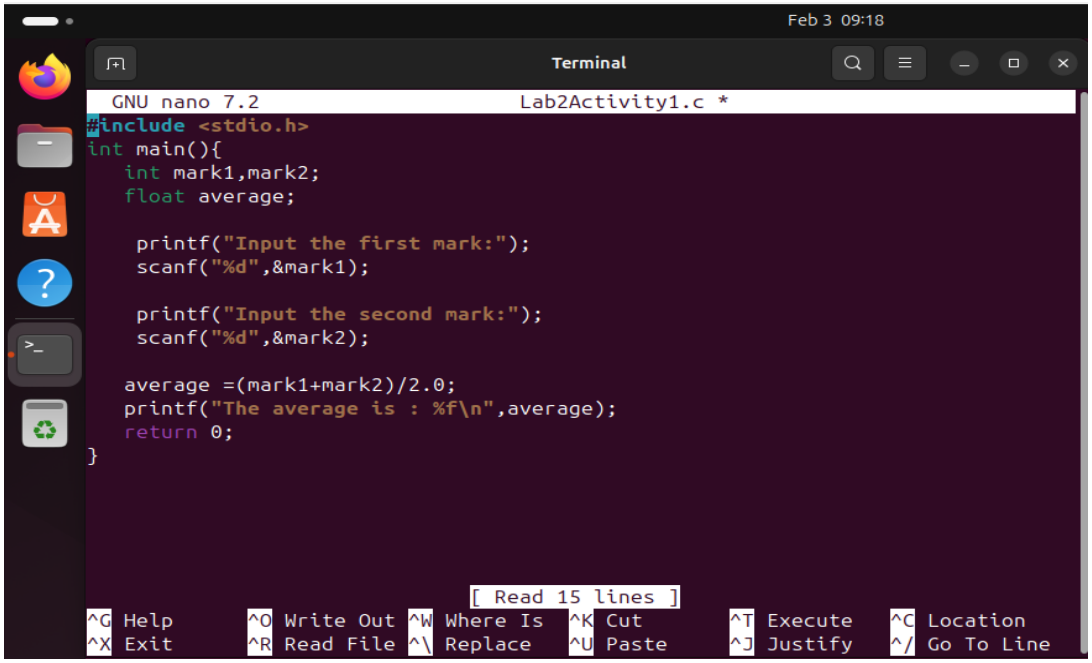
| Specialization | Data Science |
| Batch No | Y2.S2-WD-DS-0102 |

## Activity 1




This program asks the user to enter two marks, then it adds them and divides by 2.0 to calculate the average. Finally, it prints the average as a float value. Using 2.0 ensures the result includes decimals.

## Activity2

```
$ nano Lab2Activity2.c
$ gcc -oLab2Activity2 Lab2Activity2.c
$ ./Lab2Activity2
Input the first mark:70
Input the second mark:80
The average is : 75.000000
Pass
$
```

```
  GNU nano 7.2                    Lab2Activity2.c
#include <stdio.h>
int main(){
   int mark1,mark2;
   float average;

    printf("Input the first mark:");
    scanf("%d",&mark1);

    printf("Input the second mark:");
    scanf("%d",&mark2);

   average =(mark1+mark2)/2.0;
   printf("The average is : %f\n",average);

  if(average >45)
{   printf("Pass\n");
}else{
    printf("Fail\n");
}
   return 0;
                          [ Read 21 lines ]
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^/ Go To Line
```

## Activity3

```
$ nano Lab2Activity3.c
$ gcc -oLab2Activity3 Lab2Activity3.c
$ ./Lab2Activity3
Enter gender(M/F):M
Enter age:70
Senior Male Citizen
$ gcc -oLab2Activity3 Lab2Activity3.c
$ ./Lab2Activity3
Enter gender(M/F):F
Enter age:80
Senior Female Citizen
$ gcc -oLab2Activity3 Lab2Activity3.c
$ ./Lab2Activity3
Enter gender(M/F):M
Enter age:60
Not a senior Citizen
$
```

```
GNU nano 7.2                    Lab2Activity3.c
#include <stdio.h>
int main()
{
   char gender;
   int age ;

  printf("Enter gender(M/F):");
  scanf("%c", &gender);

  printf("Enter age:");
  scanf("%d", &age);

 if(age>=65)
  {
      if(gender=='M' || gender=='m')
{     printf("Senior Male Citizen\n");
  }else if(gender=='F'||gender=='f')
{    printf("Senior Female Citizen\n");
  }else{
     printf("Invalid gender input\n");
}
}else{
    printf("Not a senior Citizen\n");
}
   return 0;
}
```

## Activity4

```
$ nano Lab2Activity4.c
$ gcc -oLab2Activity4 Lab2Activity4.c
$ ./Lab2Activity4
Enter the mark:90
Grade: A
$ gcc -oLab2Activity4 Lab2Activity4.c
$ ./Lab2Activity4
Enter the mark:75
Grade: B
$ gcc -oLab2Activity4 Lab2Activity4.c
$ ./Lab2Activity4
Enter the mark:50
Grade: C
$ gcc -oLab2Activity4 Lab2Activity4.c
$ ./Lab2Activity4
Enter the mark:30
Grade: F
$ gcc -oLab2Activity4 Lab2Activity4.c
$ ./Lab2Activity4
Enter the mark:102
Invalid mark entered!
$
```

```
GNU nano 7.2                    Lab2Activity4.c
#include <stdio.h>
int main()
{
   int mark;

   printf("Enter the mark:");
   scanf("%d" , &mark);

  if(mark>=80 && mark<=100)
{
  printf("Grade: A\n");
}else if(mark>=70 && mark<80)
{
  printf("Grade: B\n");
}else if(mark>=45 && mark<70)
{
  printf("Grade: C\n");
}else if(mark<45 && mark>=0)
{
  printf("Grade: F\n");
}
else{
   printf("Invalid mark entered!\n");
}
  return 0;
}
```

## Activity5

```
bisheka@bisheka-VirtualBox:~$ nano Lab2Activity5.c
bisheka@bisheka-VirtualBox:~$ gcc -oLab2Activity5 Lab2Activity5.c
bisheka@bisheka-VirtualBox:~$ ./Lab2Activity5
Menu options :
1.Print Hello
2.Print Welcome
3.Print Goodbye
Enter your choice(1-3):1
Hello
bisheka@bisheka-VirtualBox:~$ ./Lab2Activity5
Menu options :
1.Print Hello
2.Print Welcome
3.Print Goodbye
Enter your choice(1-3):2
Welcome
bisheka@bisheka-VirtualBox:~$ ./Lab2Activity5
Menu options :
1.Print Hello
2.Print Welcome
3.Print Goodbye
Enter your choice(1-3):3
Goodbye
bisheka@bisheka-VirtualBox:~$ ./Lab2Activity5
Menu options :
1.Print Hello
2.Print Welcome
3.Print Goodbye
Enter your choice(1-3):4
Invalid choice
bisheka@bisheka-VirtualBox:~$
```

```
GNU nano 7.2                    Lab2Activity5.c
#include <stdio.h>
int main()
{
  int choice;

  printf("Menu options :\n");
  printf("1.Print Hello\n");
  printf("2.Print Welcome\n");
  printf("3.Print Goodbye\n");

  printf("Enter your choice(1-3):");
  scanf("%d",&choice);

 switch (choice)
{
   case 1:
       printf("Hello\n");
       break;

   case 2:
       printf("Welcome\n");
       break;
   case 3:
       printf("Goodbye\n");
       break;
   default:
       printf("Invalid choice\n");
}
^G Help      ^O Write Out   ^W Where Is   ^K Cut      ^T Execute
^X Exit      ^R Read File   ^\ Replace    ^U Paste    ^J Justify
```

## Activity6

```
bisheka@bisheka-VirtualBox:~$ nano Lab2Activity6.c
bisheka@bisheka-VirtualBox:~$ gcc -oLab2Activity6 Lab2Activity6.c
bisheka@bisheka-VirtualBox:~$ ./Lab2Activity6
Enter the  mark:95
Grade : A
bisheka@bisheka-VirtualBox:~$ ./Lab2Activity6
Enter the  mark:77
Grade: B
bisheka@bisheka-VirtualBox:~$ ./Lab2Activity6
Enter the  mark:55
Grade : C
bisheka@bisheka-VirtualBox:~$ ./Lab2Activity6
Enter the  mark:44
Grade : F
bisheka@bisheka-VirtualBox:~$ ./Lab2Activity6
Enter the  mark:11
Grade : F
bisheka@bisheka-VirtualBox:~$ ./Lab2Activity6
Enter the  mark:-10
Invalid mark!
bisheka@bisheka-VirtualBox:~$ ./Lab2Activity6
Enter the  mark:110
Invalid mark!
bisheka@bisheka-VirtualBox:~$
```

```
GNU nano 7.2                    Lab2Activity6.c
#include <stdio.h>
int main()
{
    int mark;
    int category;

    printf("Enter the  mark:");
    scanf("%d",&mark);

    category = mark/10;

switch (category)
{
     case 10:
     case 9:
     case 8:
        printf("Grade : A\n");
        break;
     case 7:
        printf("Grade: B\n");
        break;
     case 6:
     case 5:
     case 4:
        if(mark >= 45)
            printf("Grade : C\n");
       else
           printf("Grade : F\n");

^G Help      ^O Write Out  ^W Where Is   ^K Cut      ^T Execute
^X Exit      ^R Read File  ^\ Replace    ^U Paste    ^J Justify
```

```
switch (category)
{
     case 10:
     case 9:
     case 8:
        printf("Grade : A\n");
        break;
     case 7:
        printf("Grade: B\n");
        break;
     case 6:
     case 5:
     case 4:
        if(mark >= 45)
            printf("Grade : C\n");
       else
           printf("Grade : F\n");
           break;
     default:
        if(mark < 45  && mark>=0)
            printf("Grade : F\n");
       else
           printf("Invalid mark!\n");
}
     return 0;
}
```

^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify